

User Testing Results

1 Introduction

User testing was conducted to evaluate the usability, functionality, and overall user experience of the ToolShare web application. The goal was to observe how real users interact with the system and identify usability issues or areas for improvement.

The testing focused on two primary user roles:

- **Students** (equipment borrowers)
- **Faculty/Staff** (equipment providers and administrators)

This testing aligns with core system features such as booking, approval workflows, and equipment management as defined in the project scope .

2 Objectives

The objectives of user testing were:

- Evaluate ease of navigation
- Identify usability issues in workflows
- Assess clarity of tool information
- Validate checkout, return, and image upload features
- Gather actionable user feedback

3 Testing Methodology

Participants

- 3 student users
- 2 users simulating faculty/admin roles

Approach

A **task-based usability testing** method was used.

Participants were asked to complete realistic tasks without step-by-step guidance.

The facilitator:

- Observed behavior

- Recorded difficulties
- Provided help only when necessary

4 Test Scenarios & Tasks

Student Workflow

- Create account and log in
- Browse tools and add multiple items to cart
- Complete checkout
- Check booking status
- Log out

Faculty Workflow

- Log in as faculty
- Review borrow requests
- Approve one and deny one request

Tool Management

- Add, edit, and delete a tool

Checkout & Return

- Check out approved item
- Upload condition image (before)
- Return item
- Upload condition image (after)

Booking History

- View booking history
- Log out

5 Observations

- Some users hesitated when locating checkout and return actions
- Role switching (student → faculty) required adjustment
- Condition image upload worked but lacked visibility
- Overall navigation improved after initial use

6 User Responses

The following are representative responses collected during testing:

User 1 (Student)

- **UI Feedback:**
“Everything looks consistent and professional.”
- **Usability:**
“Uploading images worked, but I wasn’t sure if it uploaded successfully.”
- **Information:**
“Would like to see more info about tool usage.”
- **Overall Experience:**
“Smooth experience overall.”
- **Suggested Change:**
“Add confirmation messages after actions.”

User 2 (Student)

- **UI Feedback:**
“The design is simple and easy to understand.”
- **Usability:**
“The process was mostly smooth, but I got confused when switching between pages.”
- **Information:**
“It would be helpful to see more details about availability and restrictions.”
- **Overall Experience:**
“Pretty easy to use overall.”
- **Suggested Change:**
“Add tooltips or instructions for first-time users.”

User 3 (Both Roles)

- **UI Feedback:**
“Looks good”
- **Usability:**
“Works mostly fine”
- **Information:**
“The descriptions are too short”
- **Overall Experience:**
“Pretty good”
- **Suggested Change:**

“Make the times go from the browsing part to the shopping cart part so I don’t have to put them in over and over”

User 4 (Both Roles)

- **UI Feedback:**
“Looks really good. I like the colors.”
- **Usability:**
“It works pretty well. It’s good.”
- **Information:**
“Looks like you’d have everything you need”
- **Overall Experience:**
“It seems like it works good”
- **Suggested Change:**
“Nothing, it looks good to me”

User 5 (Faculty Role)

- **UI Feedback:**
“UI looks really good”
- **Usability:**
“Works for what I would need”
- **Information:**
“Shows what we I need”
- **Overall Experience:**
“Pretty good”
- **Suggested Change:**
“Nothing really”

7 User Feedback Summary

UI & Design

- Clean and modern layout overall
- Minor spacing and alignment inconsistencies

Usability

- Checkout and return flow slightly confusing initially
- Some buttons not immediately noticeable
- Annoyed that available time checked while browsing did not carry over to cart

Information Clarity

- Tool details were useful but could be more detailed

Overall Experience

- Generally smooth and intuitive after short learning curve

8 Conclusion

User testing confirmed that ToolShare effectively supports its core functionality of equipment sharing between students and faculty.

Key strengths:

- Clean interface
- Functional booking system
- Clear role-based workflows

Areas for improvement:

- Visibility of key actions (checkout, return, approval)
- Better feedback for user actions (uploads, confirmations)
- Enhanced clarity for first-time users

Overall, the system provides a strong user experience and meets its intended goals, with minor UI/UX improvements recommended.